The convergence tests that we have looked at so far apply only to series with positive terms. We now turn to series whose terms are not necessarily positive. Of particular importance are *alternating series*, whose terms alternate in sign.

1 Alternating Series

Definition 1.1 (Alternating series)**.** An *alternating series* is a series $\sum_{n=1}^{\infty} a_n$ where a_n is of the form $a_n = (-1)^{n-1} b_n$ or $a_n = (-1)^n b_n$, where $b_n > 0$.

1. Show that $b_n = |a_n|$ in the definition of an alternating series.

Example 1.2. (a) $\displaystyle\sum_{n=1}^{\infty} (-1)^{n-1} \frac{1}{n} = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \frac{1}{6} + \cdots.$

(b) $\displaystyle\sum_{n=1}^{\infty} (-1)^n \frac{n}{n+1} = -\frac{1}{2} + \frac{2}{3} - \frac{3}{4} + \frac{4}{5} - \frac{5}{6} + \frac{6}{7} - \cdots$

The following test says that if the terms of an alternating series decrease toward 0 in absolute value, then the series converges.

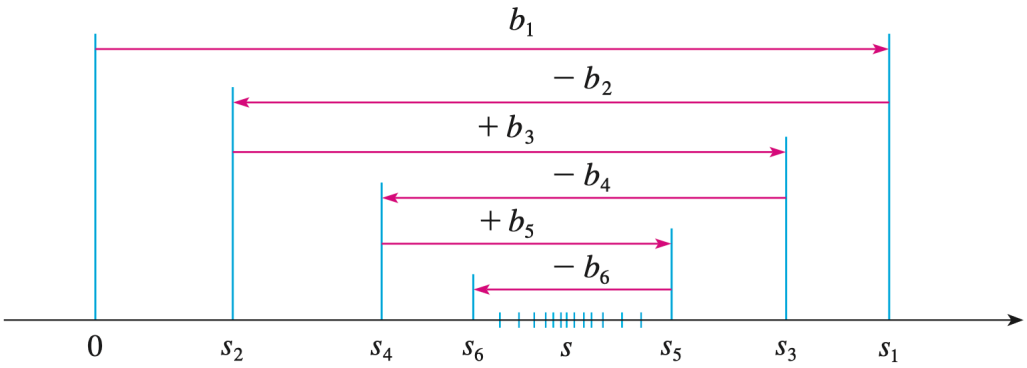
Theorem 1.3 (Alternating Series Test)**.** *If the alternating series* $\sum_{n=1}^{\infty} (-1)^{n-1} b_n$ *satisfies*

(i) $b_{n+1} \le b_n$ *for all* n*, and*

(ii) $\lim_{n\to\infty} b_n = 0$*,*

then the series is convergent.

Such an alternating series is shown below. Notice that the even partial sums are increasing, while the odd partial sums are decreasing. It is therefore plausible that the sequences $\{s_{2k}\}$ and $\{s_{2k+1}\}$ both converge to a common value s, so that $\lim_{k\to\infty} s_k = s$.



First, we need to actually prove that if $\lim_{n\to\infty} s_{2n} \to s$ and $\lim_{n\to\infty} s_{2n+1} \to s$, then $\lim_{n\to\infty} s_n \to s$.

2. Prove this.

Proof. Consider the sequence of even partial sums. Letting $s_0 := 0$, for $n \geq 1$ we can write

$$s_{2n} = s_{2n-2} + (b_{2n-1} - b_{2n}).$$

By (i), $b_{2n} \leq b_{2n-1}$ and therefore $b_{2n-1} - b_{2n} \geq 0$. Hence, $s_{2n} \geq s_{2n-2}$. This shows that the sequence $\{s_{2n}\}$ is increasing. We can also write

$$s_{2n} = \sum_{k=1}^{2n} (-1)^{k-1} b_k = b_1 - \sum_{k=1}^{n-1} (b_{2k} - b_{2k+1}) - b_{2n}$$

Since $(b_{2k} - b_{2k+1}) \geq 0$ for all $k = 1, \ldots, n-1$, this shows that $s_{2n} \leq b_1$ for all n; that is, the sequence $\{s_{2n}\}$ is bounded above. Since $\{s_{2n}\}$ is increasing and bounded above, it converges by the Bounded Monotone Convergence Theorem. Let $s := \lim_{n\to\infty} s_{2n}$.

3. Now write $s_{2n+1} = s_{2n} + b_{2n+1}$ and compute $\lim_{n\to\infty} s_{2n+1}$ to complete the proof.

□

2 Exercises

4. We have seen that the harmonic series $\sum_{n=1}^{\infty} \frac{1}{n}$ diverges. Use the Alternating Series Test to show that that *alternating harmonic series* $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n}$ converges.

5. Determine whether $\sum_{n=1}^{\infty} \frac{(-1)^n 3n}{4n-1}$ converges or diverges.

6. Determine whether $\sum_{n=1}^{\infty} \frac{(-1)^{n+1} n^2}{n^3+1}$ converges or diverges. (*Hint*: To check condition (i) of the Alternating Series Test, consider the differentiable function $f(x) = \frac{x^2}{x^3+1}$.)

3 Estimating Sums

The next theorem says that for series that satisfy the conditions of the Alternating Series Test, the size of the error $s - s_n$ in approximating the sum of the series s by its nth partial sum s_n is smaller than b_{n+1}, which is the absolute value of the first neglected term.

Theorem 3.1 (Alternating Series Estimation Theorem)**.** *If* $s = \sum_{n=1}^{\infty} (-1)^{n-1} b_n$ *is the sum of an alternating series that satisfies the conditions of the Alternating Series Test, then*

$$|R_n| = |s - s_n| \leq b_{n+1}.$$

Proof. We showed in the proof of the Alternating Series Test that $s_{2n} \leq s$ for all n. A similar argument shows that $s \leq s_{2n+1}$ for all n. This shows that s is between any two consecutive partial sums; that is, $s_n \leq s \leq s_{n+1}$ if n is odd, and $s_{n+1} \leq s \leq s_n$ if n is even.

7. Noting that $|s_{n+1} - s_n| = b_{n+1}$, show that it follows from the above that $|R_n| = |s - s_n| \leq b_{n+1}$.

□

7. Find the sum of the series $\sum_{n=0}^{\infty} \frac{(-1)^n}{n!}$ correct to three decimal places.